Basic Usage 2

Read data from soil moisture sensor

This is the basic experiment for learning the plant watering kit, it will help you to know about the pinout of electronics components onboard and help you understand how to use those components individual.

OK, let’s begin the basic learning.

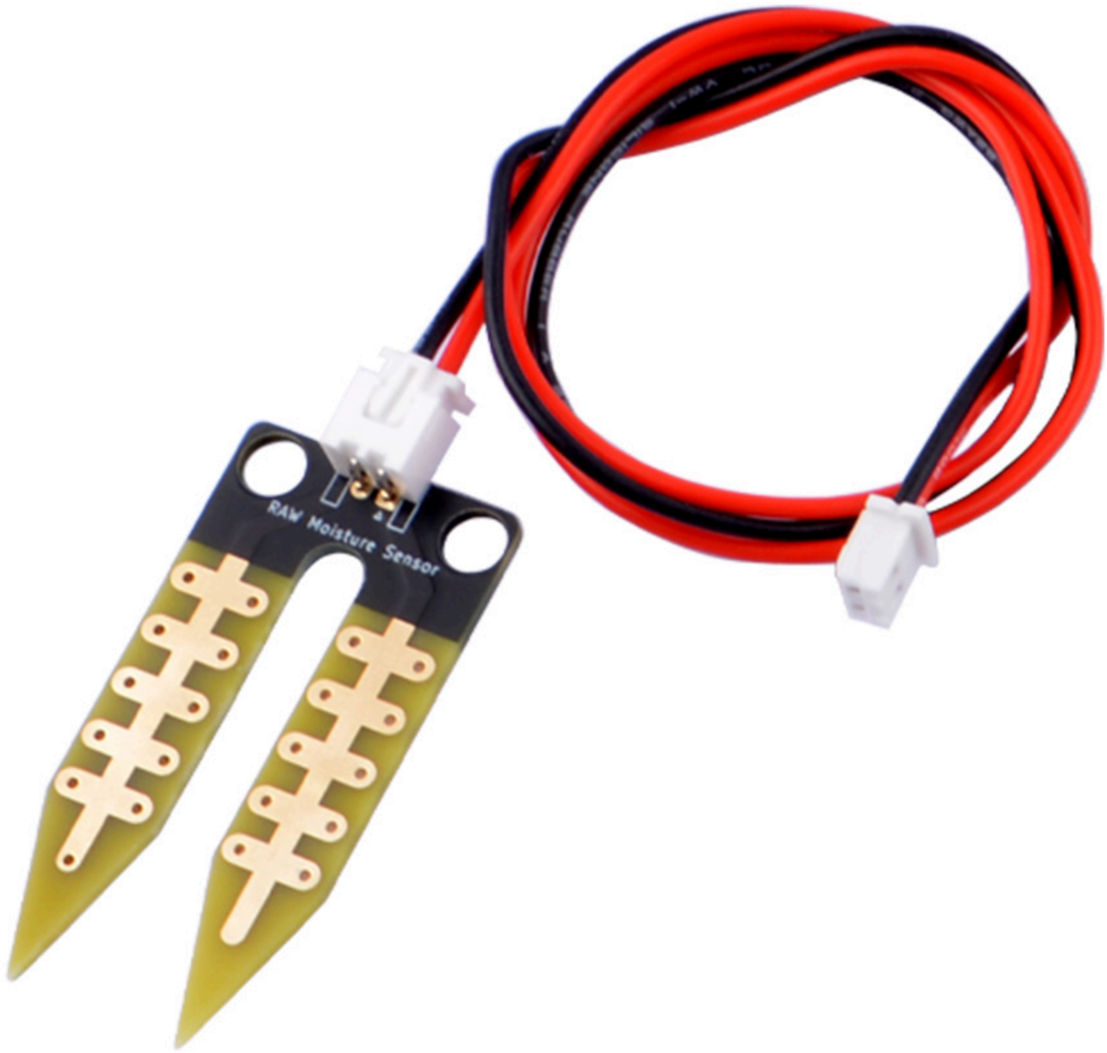
In this experiment, we are going to read raw data from soil moisture sensors on the plant watering hat board.

Hardware Overview

The shield provides the following interfaces:

- 3 x Soil Moisture Sensors (Analog Inputs A0, A1, A2)
- 3 x NTC Temperature Sensors (Analog Inputs A4, A5, A6)
- 3 x 3.3V Relay Modules (Digital Outputs 2, 3, 4)
- 3 x Mini Water Pumps
- 1.3-inch IPS RGB TFT Screen (ST7789 Controller)

Soil Moisture sensor Details



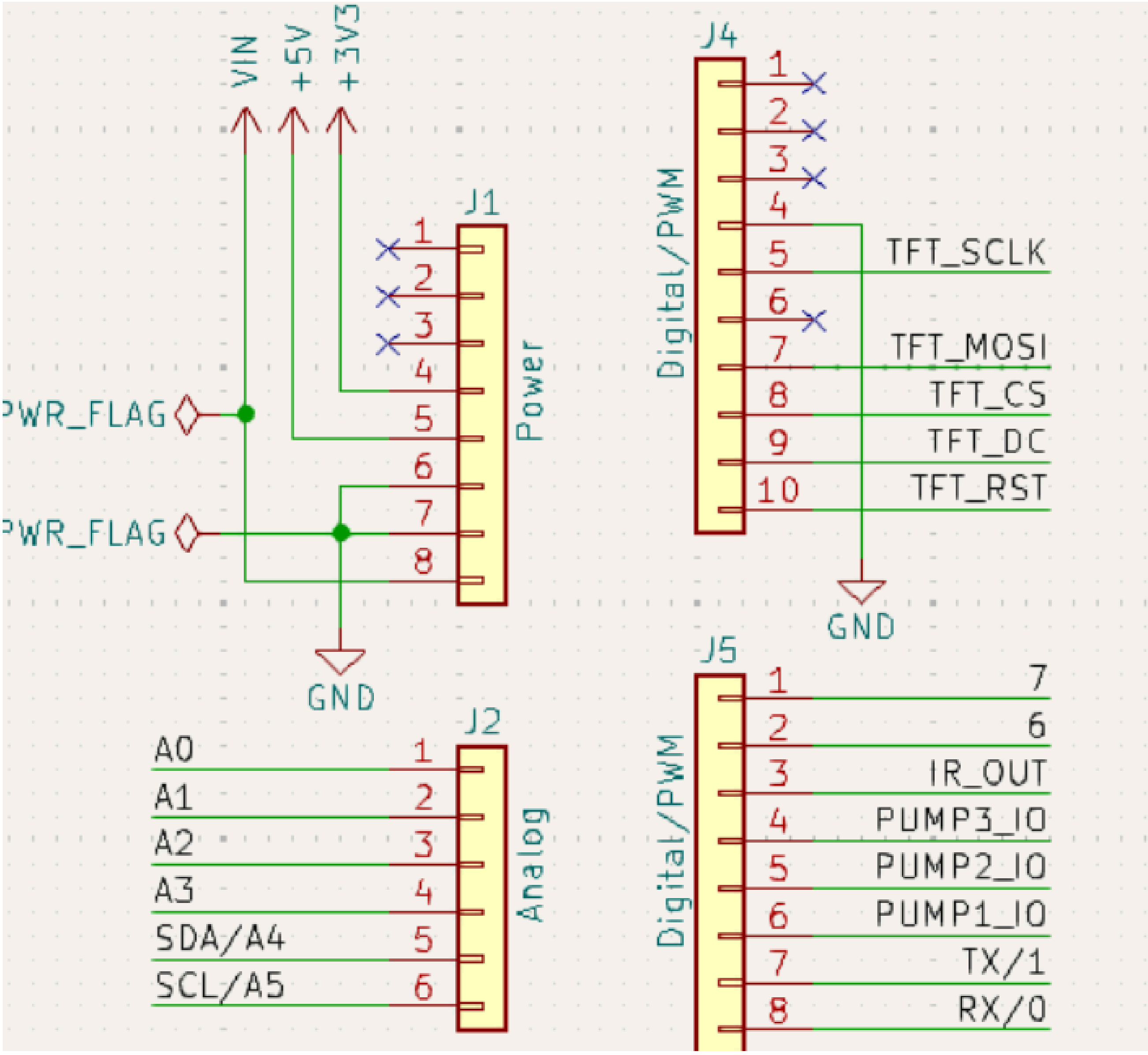
Pinout Chart

- Details of the expansion board.

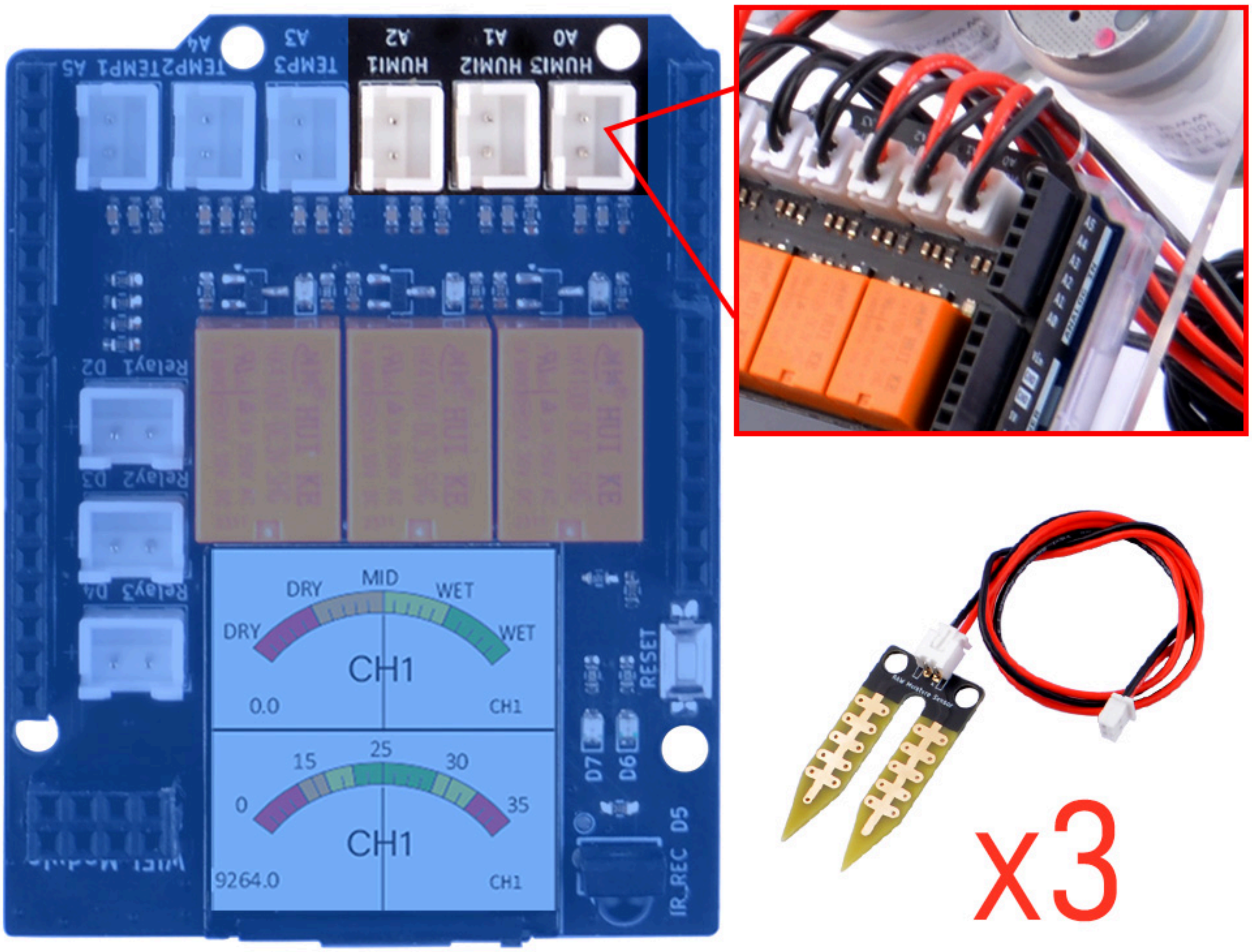
Plant Watering Kit Hat Board	Arduino UNO R4 WiFi Board
HUMI3	A0
HUMI2	A1
HUMI1	A2
TEMP3	A3
TEMP2	A4
TEMP1	A5

Plant Watering Kit Hat Board	Arduino UNO R4 WiFi Board
IR_RSV	D5
Relay 1	D2
Relay 2	D3
Relay 3	D4
TFT_SCLK	D13
TFT_MOSI	D11
TFT_CS	D10
TFT_DC	D9
TFT_RST	D8
RX	TX->1
TX	RX<-0
Green LED	D6
Red LED	D7

Circuit Diagram



Soil moistrue socket position

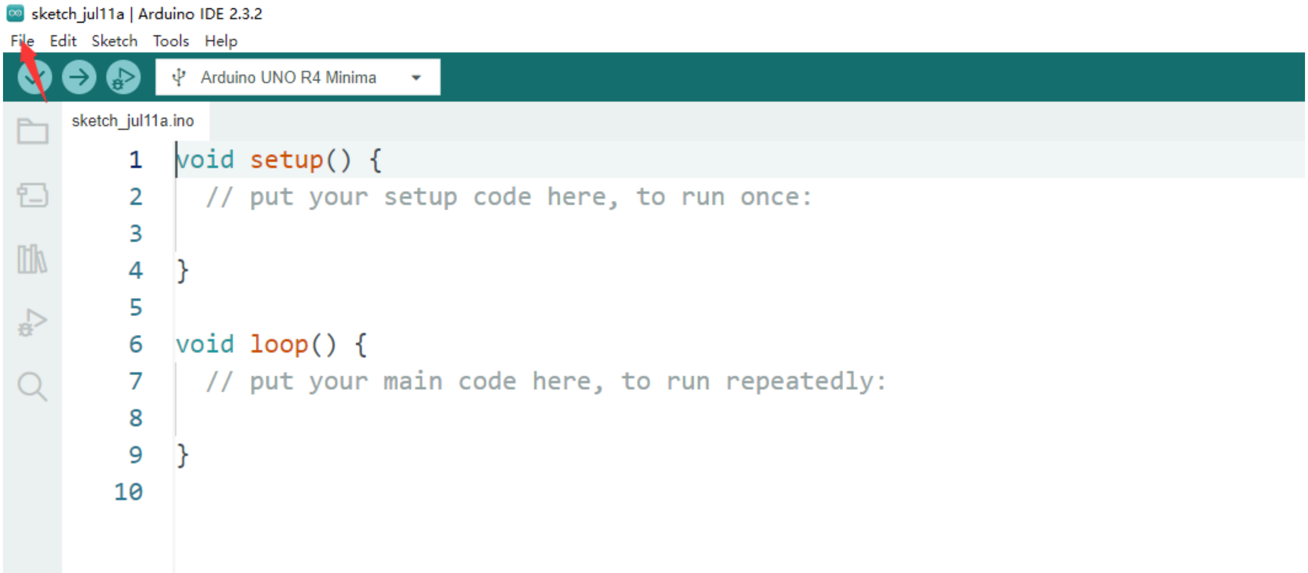


Connecting the Shield

- place the arduino uno r4 on a flat surface.
- align the shield with the headers of the arduino board and gently press it down until it clicks into place.
- plug the plant watering hat board on top of arduino uno r4 on gpio pins.

Programming

Open arduino IDE and create a new sketch by clicking `file` -> `New Sketch`



Define LED Pin number

```
#define HUMI3 A0
#define HUMI2 A1
#define HUMI1 A2
```

Initializing Pin and Serial port for monitoring.

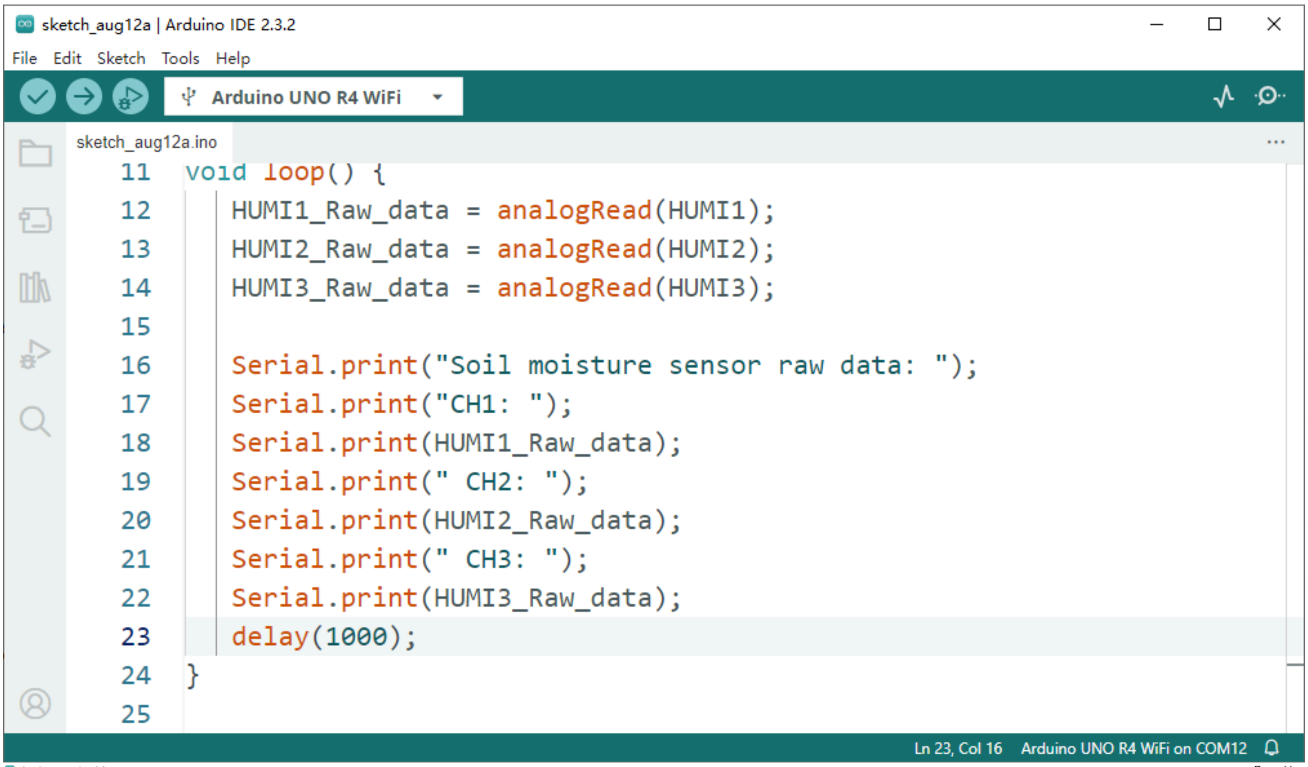
```
void setup() {
    Serial.begin(9600);  // Initializing Serial monitor
}
```

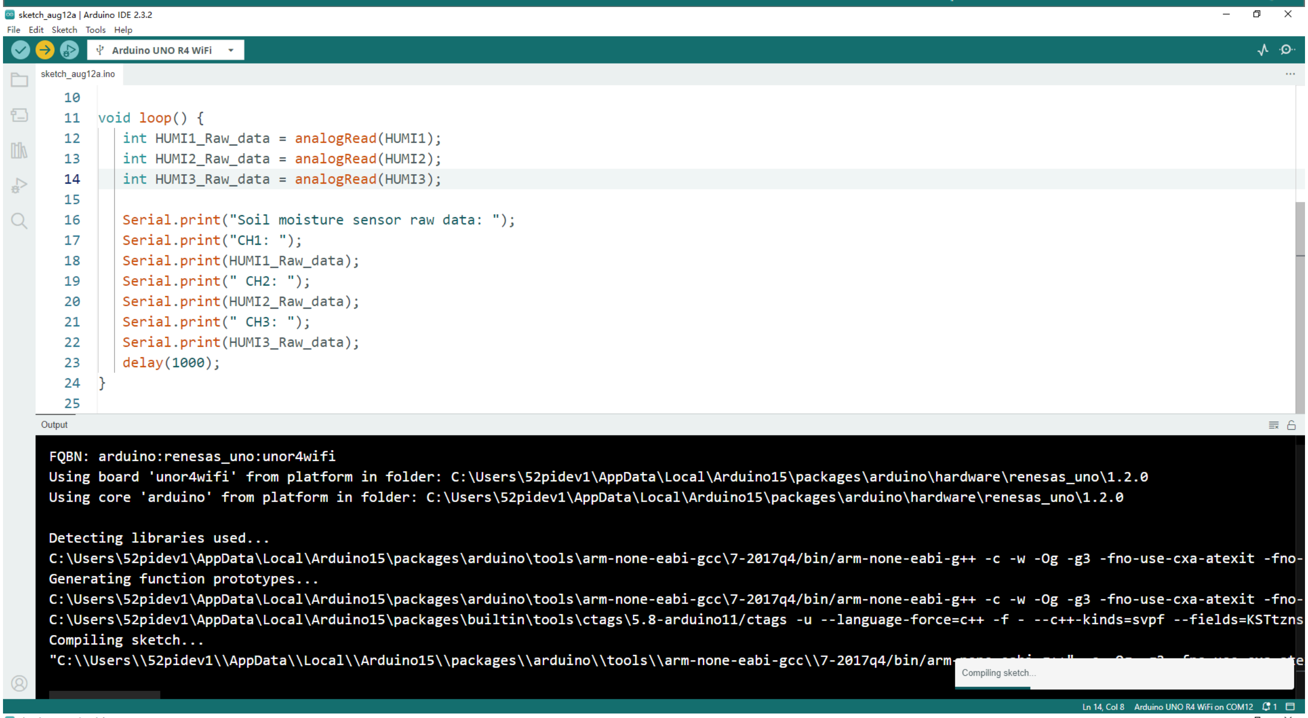
Modify loop section

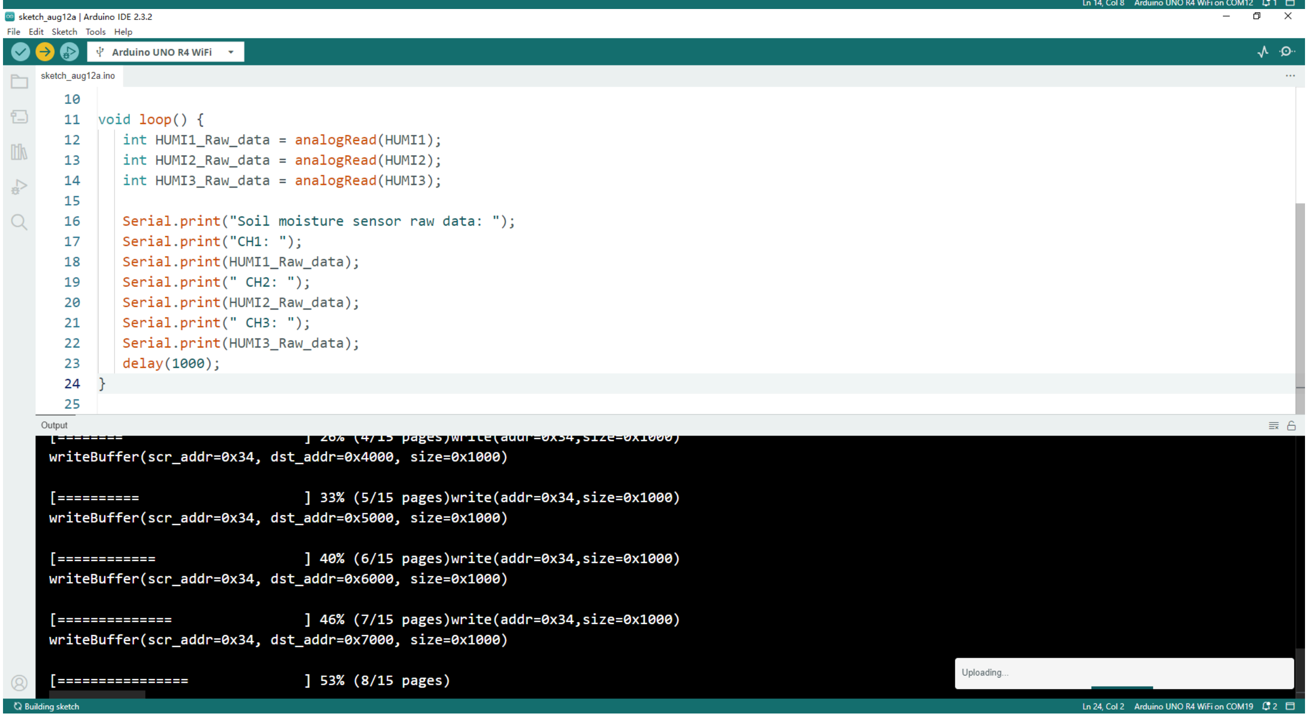
```
void loop() {
    int HUMI1_Raw_data = analogRead(HUMI1);
    int HUMI2_Raw_data = analogRead(HUMI2);
    int HUMI3_Raw_data = analogRead(HUMI3);

    Serial.print("Soil Moisture sensor raw data:");
    Serial.print("CH1: ");
    Serial.print(HUMI1_Raw_data);
    Serial.print("CH2: ");
    Serial.print(HUMI2_Raw_data);
    Serial.print("CH3: ");
    Serial.println(HUMI3_Raw_data);
    delay(1000); //wait for a second.
}
```

Upload the sketch to Arduino UNO R4 WiFi board.

- Connect the Arduino UNO R4 WiFi board to your computer via USB-C cable on USB port
- Select the serial device on your arduino IDE and click upload icon as following figure:

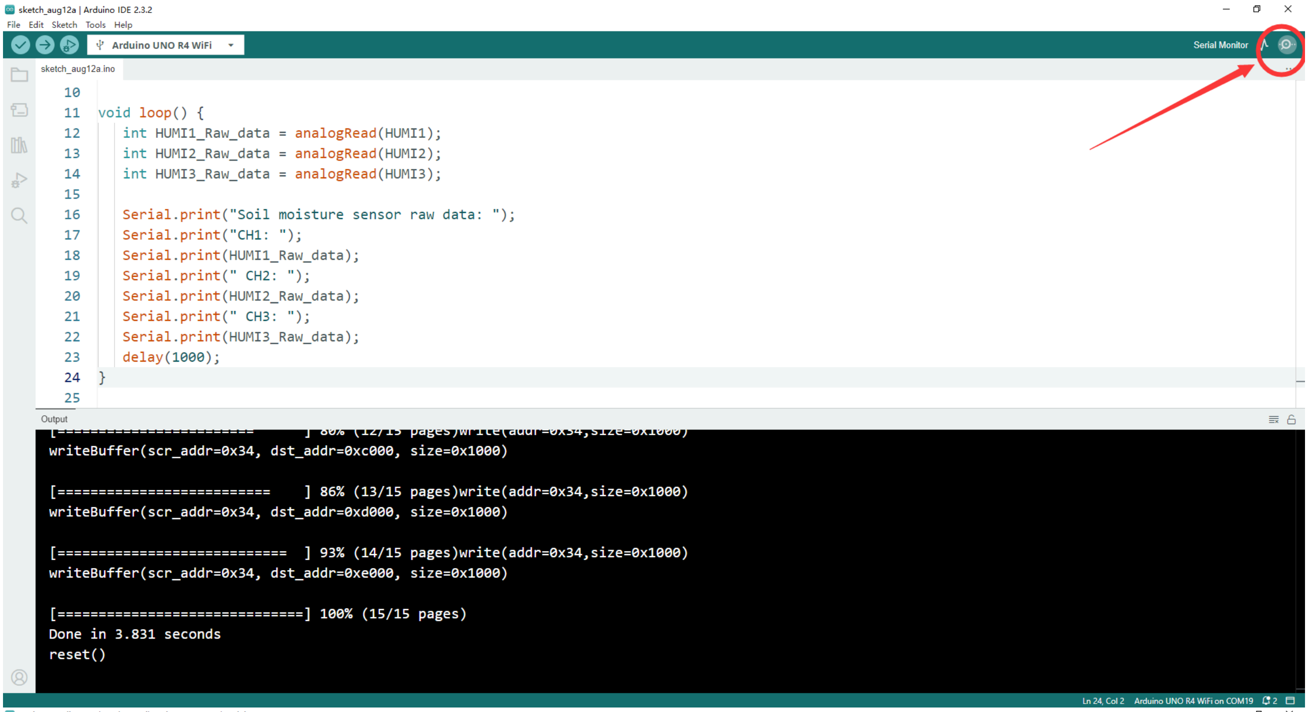

sketch_aug12a | Arduino IDE 2.3.2
File Edit Sketch Tools Help
Arduino UNO R4 WiFi
sketch_aug12a.ino
11 void loop() {
12 HUMI1_Raw_data = analogRead(HUMI1);
13 HUMI2_Raw_data = analogRead(HUMI2);
14 HUMI3_Raw_data = analogRead(HUMI3);
15
16 Serial.print("Soil moisture sensor raw data: ");
17 Serial.print("CH1: ");
18 Serial.print(HUMI1_Raw_data);
19 Serial.print(" CH2: ");
20 Serial.print(HUMI2_Raw_data);
21 Serial.print(" CH3: ");
22 Serial.print(HUMI3_Raw_data);
23 delay(1000);
24 }
25
Ln 23, Col 16 Arduino UNO R4 WiFi on COM12

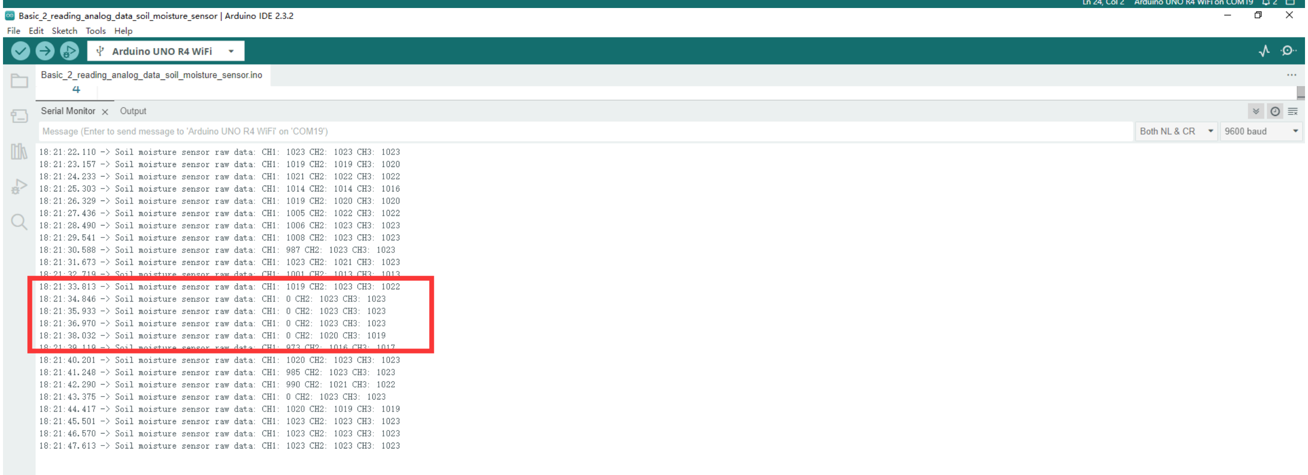

sketch_aug12a | Arduino IDE 2.3.2
File Edit Sketch Tools Help
Arduino UNO R4 WiFi
sketch_aug12a.ino
10
11 void loop() {
12 int HUMI1_Raw_data = analogRead(HUMI1);
13 int HUMI2_Raw_data = analogRead(HUMI2);
14 int HUMI3_Raw_data = analogRead(HUMI3);
15
16 Serial.print("Soil moisture sensor raw data: ");
17 Serial.print("CH1: ");
18 Serial.print(HUMI1_Raw_data);
19 Serial.print(" CH2: ");
20 Serial.print(HUMI2_Raw_data);
21 Serial.print(" CH3: ");
22 Serial.print(HUMI3_Raw_data);
23 delay(1000);
24 }
25
Output
FQBN: arduino:renesas_uno:unor4wifi
Using board 'unor4wifi' from platform in folder: C:\Users\52pidev1\AppData\Local\Arduino15\packages\arduino\hardware\renesas_uno\1.2.0
Using core 'arduino' from platform in folder: C:\Users\52pidev1\AppData\Local\Arduino15\packages\arduino\hardware\renesas_uno\1.2.0
Detecting libraries used...
C:\Users\52pidev1\AppData\Local\Arduino15\packages\arduino\tools\arm-none-eabi-gcc\7-2017q4/bin/arm-none-eabi-g++ -c -w -Og -g3 -fno-use-cxa-atexit -fno-
Generating function prototypes...
C:\Users\52pidev1\AppData\Local\Arduino15\packages\arduino\tools\arm-none-eabi-gcc\7-2017q4/bin/arm-none-eabi-g++ -c -w -Og -g3 -fno-use-cxa-atexit -fno-
C:\Users\52pidev1\AppData\Local\Arduino15\packages\builtin\tools\ctags\5.8-arduino11/ctags -u --language-force=c++ -f - --c++-kinds=svpf --fields=KSTtzns
Compiling sketch...
"C:\\Users\\52pidev1\\AppData\\Local\\Arduino15\\packages\\arduino\\tools\\arm-none-eabi-gcc\\7-2017q4/bin/arm
Compiling sketch...
Ln 14, Col 8 Arduino UNO R4 WiFi on COM12


sketch_aug12a | Arduino IDE 2.3.2
File Edit Sketch Tools Help
Arduino UNO R4 WiFi
sketch_aug12a.ino
10
11 void loop() {
12 int HUMI1_Raw_data = analogRead(HUMI1);
13 int HUMI2_Raw_data = analogRead(HUMI2);
14 int HUMI3_Raw_data = analogRead(HUMI3);
15
16 Serial.print("Soil moisture sensor raw data: ");
17 Serial.print("CH1: ");
18 Serial.print(HUMI1_Raw_data);
19 Serial.print(" CH2: ");
20 Serial.print(HUMI2_Raw_data);
21 Serial.print(" CH3: ");
22 Serial.print(HUMI3_Raw_data);
23 delay(1000);
24 }
25
Output
writeBuffer(scr_addr=0x34, dst_addr=0x4000, size=0x1000)
[=========] 33% (5/15 pages)write(addr=0x34,size=0x1000)
writeBuffer(scr_addr=0x34, dst_addr=0x5000, size=0x1000)
[===========] 40% (6/15 pages)write(addr=0x34,size=0x1000)
writeBuffer(scr_addr=0x34, dst_addr=0x6000, size=0x1000)
[=============] 46% (7/15 pages)write(addr=0x34,size=0x1000)
writeBuffer(scr_addr=0x34, dst_addr=0x7000, size=0x1000)
[===============] 53% (8/15 pages)
Uploading...
Building sketch
Ln 24, Col 2 Arduino UNO R4 WiFi on COM19

Open Serial monitor

- Click this icon on right up cornner to open `serial monitor`.





Demo Code Sketch Download

[Basic_2_demo_code_sketch_download]()

Finally

- Check if the three channel has outputs on the serial monitor, and try to put one of them into the water, and observe the output data on screen. If you can see the data changed when you insert the sensor into water, it means you finished this task. Let us remove to next chapter.